Project Design Phase-II
Data Flow Diagram & User Stories

Date	23 March 2026
Team ID	69c0f3744241397902eb6826
Project Name	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
Maximum Marks	4 Marks

◈ Data Flow Diagram (DFD)

Level 0 – Context Diagram

Actors:

- User / Admin
- External Data Source (CSV, Excel, APIs)

Process:

- Data Import Automation System

Flow:

- User uploads data → System processes → Data stored in database → Output reports generated

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Level 1 – Detailed DFD

Processes:

1. Data Upload
2. Import Set Creation
3. Transform Map Execution
4. Relationship Mapping (Dot Walking)

5. Data Storage

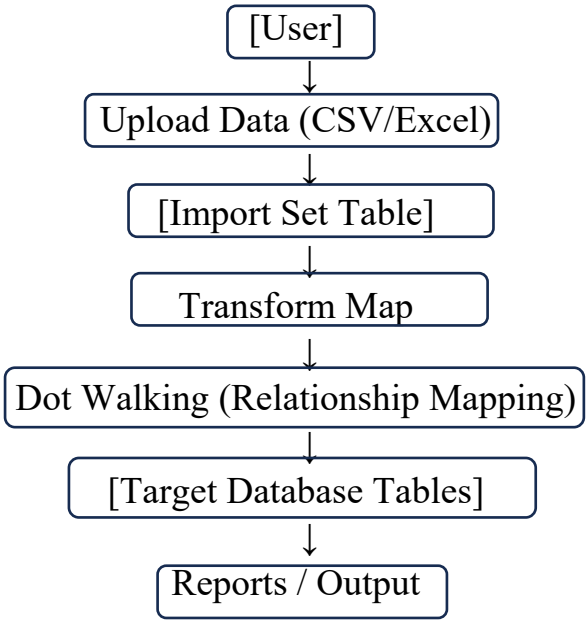
Data Stores:

- Import Set Table
- Target Tables (e.g., User, Company, Asset)

Flow Explanation:

- External file → Uploaded by user
- Stored in **Import Set Table**
- Transform Map processes data
- Dot Walking links related tables (e.g., user.company.name)
- Final structured data stored in Target Tables

Simple Text Representation of DFD



◈ User Stories

1. Admin User Story

- As an **admin**, I want to upload external data files so that I can import records into the system efficiently.

2. Data Analyst User Story

- As a **data analyst**, I want automatic field mapping using Transform Maps so that manual work is reduced.

3. System User Story

- As a **system**, I want to validate incoming data so that errors are minimized during import.

4. Developer User Story

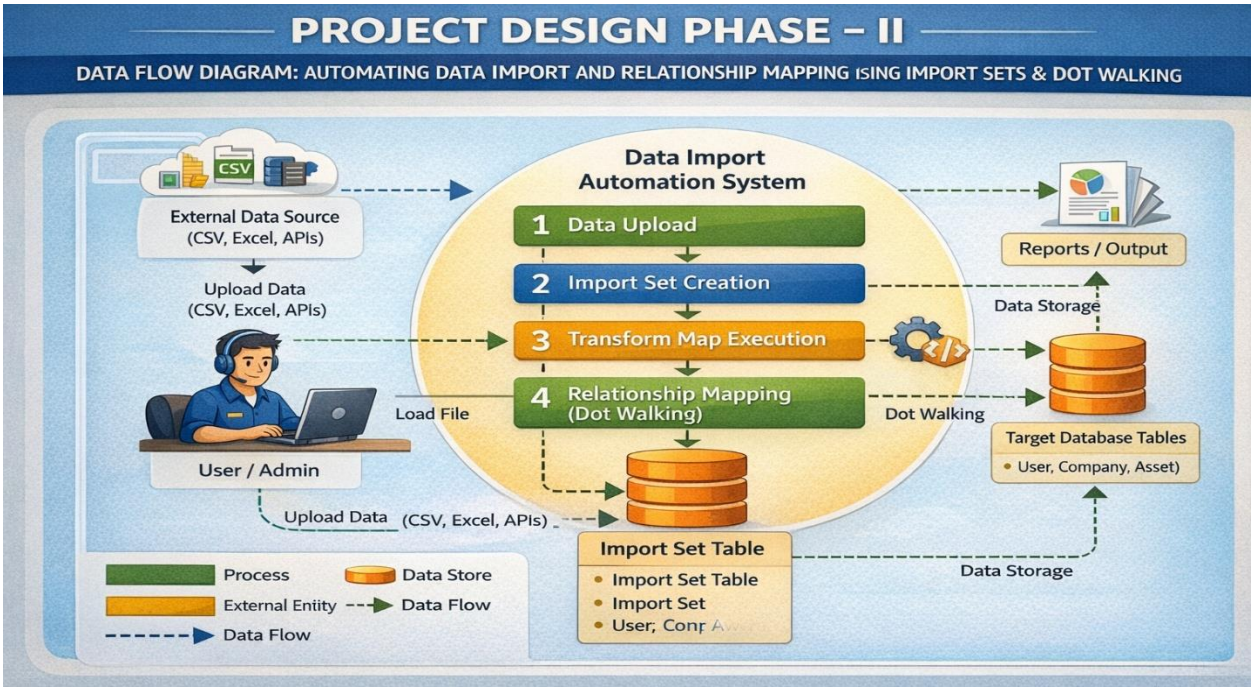
- As a **developer**, I want to use Dot Walking to map relationships between tables so that data linking is accurate.

5. End User Story

- As an **end user**, I want clean and structured data so that I can generate accurate reports.

◈ Acceptance Criteria

- Data should be imported without manual intervention
- Relationships should be correctly mapped using Dot Walking
- Errors should be logged and displayed
- Data integrity must be maintained



◈ Expected Outcome

- Faster data import process
- Reduced manual errors
- Efficient relationship mapping
- Improved reporting accuracy